

Generative AI-Powered Conversational Assistant for Document, Web, and Repository Interaction and Retrieval

Shailaja Uke

Department of Computer Engineering
Vishwakarma Institute of technology
Pune, India
shailaja.uke@vit.edu

Ayush Laddha

Department of Computer Engineering
Vishwakarma Institute of technology
Pune, India
laddha.ayush22@vit.edu

Shreya Bambal

Department of Computer Engineering
Vishwakarma Institute of technology
Pune, India
shreya.bambal22@vit.edu

Atharva Bonde

Department of Computer Engineering
Vishwakarma Institute of technology
Pune, India
atharva.bonde22@vit.edu

Prasanna Atram

Department of Computer Engineering
Vishwakarma Institute of technology
Pune, India
prasanna.atram22@vit.edu

Abstract— With continuously increasing digital content and resources, it has become even more essential to have simple, intuitive means of retrieving and understanding information. This paper develops a new AI-powered system that enables interactive learning by three different features: "Chat with PDF", "Chat with Repository", and "Chat with URL". Based on state-of-the-art language models and retrieval-augmented generation techniques, it allows for the extraction, querying, and understanding of all the information that comes from the most varied sources with even less effort. Advanced vector stores, conversational retrieval chains and document embedding strategies form the architecture to ensure high-accuracy relevant responses. The paper emphasises system design, implementation challenges as well as its practical use in education, software development, and research. The results of evaluating this system show a substantial increase in user engagement as well as knowledge acquisition. Our work is meant to fill the gap between statically consumed content and the dynamically intelligent AI-driven interface-a new generation of very useful study aids.

Keywords—Retrieval-Augmented Generation (RAG), Vector Stores, AI-Powered Learning, Conversational AI, Information Retrieval

I. INTRODUCTION

There has been tremendous change in how humans engage with, interact with, and also understand digital content in recent years, particularly regarding natural language processing and machine learning. While good for increasing online documents, inter alia, web pages, PDFs, and code repositories are only in greater demand from such data for meaningful insights through information extraction tools. This project is designed to create an intelligent system with which users interact using natural-language queries whereby they can interact with diverse types of digital information. In this regard, the system assists users to "chat" with many sources of information such as web URLs or PDFs and other GitHub repository information for relevancy and context-awareness in the response generated.

These applied leading-edge technologies, such as conversational AI, which is part of the LangChain framework, document vector search on Qdrant, and Groq for higher AI model performance. On top of this, embeddings produced by advanced models, such as BGE (Bayesian Generative

Embeddings) from HuggingFace, are applied in representing textual information to be usable in semantic searches and retrievals. This technique allows the system to provide a better version of the Retained-Augmented Generation (RAG) model to answer queries based on information retrieval and generative models accurately.

This tool is highly additive for learning and research purposes as it offers users a conversational way of accessing and understanding complex documents. It would, therefore, prove to be an indispensable resource for students, researchers, and developers alike. The project also demonstrates how conversational AI may be easily incorporated with document databases to make information retrieval more accessible, intuitive, and effective.

The study investigated how NLP, document retrieval, and AI-powered generation could be integrated through this research study to provide a fluid user experience. This research contributed to the pool of literature on AI-driven systems for document interaction in both academic and professional contexts.

II. RELATED WORK

The paper reviews RankAE's role in summarising chat logs using a topic-oriented ranking strategy to capture key utterances based on centrality and diversity. A denoising auto-encoder (DAE) refines these summaries by eliminating redundancies while preserving context, producing concise outputs. Its unsupervised approach allows for scalable application on large datasets. RankAE outperforms techniques like TextRank, offering a balance between relevance and brevity, making it particularly effective for customer service logs.[1]

This review covers a Python-based PDF tool with CustomTkinter for offline PDF manipulation, offering split, merge, rotate, and encrypt functions without internet dependency. The customizable interface ensures intuitive use, enhancing document management efficiency. By providing comprehensive offline capabilities, this tool offers flexible, secure, and user-friendly solutions for PDF tasks.[2]

The paper discusses a chatbot using annotated movie dialogues and a Transformer-based model for generating

relevant, engaging responses. By fine-tuning domain-specific data, the chatbot improves intent prediction and knowledge retrieval, surpassing commercial alternatives in conversational flow and accuracy, illustrating its role in advancing specialised conversational AI.[3]

The paper showcases a chatbot using tokenized PDF content, vector embeddings, and cosine similarity for accurate academic queries. Retrieval-Augmented Generation (RAG) ensures context-aware responses, with multi-PDF querying capabilities enhancing complex research tasks. The review highlights its effective use of LLMs and RAG to transform document interaction.[4]

This paper explores a system generating question-answer pairs to evaluate RAG-based LLMs in the medical field, focusing on factoid, long-form, and unanswerable questions to assess model accuracy. Using validated hospital datasets, it efficiently evaluates chatbot performance. Testing with Llama-2-13B demonstrated effective question-answer differentiation, enhancing QA evaluation frameworks.[5]

This paper looks at a chatbot that mixes sentiment analysis and graph-based interactions for better customer service conversations. It manages conversations smoothly and provides an easy-to-use admin interface with Vue.js and Vuex. By using Russell's Model for sentiment analysis, it boosts customer engagement and makes service more efficient.[6]

The focus here is on using LLMs (Large Language Models) for data analysis. It simplifies coding and tackles issues like hallucinations and high costs. Methods like Retrieval-Augmented Generation (RAG) and vector databases improve the accuracy of responses. Plus, breaking down complex queries into smaller tasks makes everything faster and more precise.[7]

Commissioned by Innovation, this project created a chatbot using internal documents to provide multilingual responses. It ensures secure data access and uses Swiss infrastructure for sustainability. The chatbot uses RAG for precise answers and has a user-friendly interface, making document-based interactions more flexible and reliable.[8]

This study looks at integrating LLMs with formats like PDFs using frameworks like LangChain. RAG and vector embeddings improve how data is retrieved and analysed, making the system more scalable, efficient, and secure for multilingual applications. It's all about improving AI-driven user interactions.[9]

This project uses OpenAI Codex and internal documents for a chatbot that supports multiple languages and secure Swiss deployment. Emphasising RAG-based responses, intuitive user interfaces, and minimal carbon footprints, it provides reliable, industry-relevant content, underscoring LLM potential in education with quality oversight.[10]

This study analyses trends and challenges in Pre-Trained Model reuse on HuggingFace, including guidance gaps, comprehension issues, and documentation limitations. Persistent reuse issues and evolving trends are tracked, offering insights and recommendations to enhance PTM usability in software engineering and deep learning communities.[11]

This review highlights how large generative models like ChatGPT and DALL-E-2 have transformed sectors through tasks such as text-to-image and text-to-code generation.

Generative AI impacts industries by enhancing creativity and productivity through novel content creation using vast datasets and high-dimensional data mapping. Advances in neural networks and transformers enable these models to handle complex data and produce diverse outputs. The paper underscores its influence in areas like art and education, where generative AI complements human creativity. It also categorises models based on input-output mappings and explores their practical applications, emphasising their transformative real-world potential.[12]

The review explores how Large Language Models (LLMs) like ChatGPT can transform user interactions with PDF-based automotive manuals, facilitating effective and simplified query resolution. Traditional manual-based troubleshooting is often inefficient, leading to user frustration. By leveraging LLMs, users can describe issues in natural language, and AI-powered assistants provide quick and relevant responses. This integration improves the accessibility and understanding of complex vehicle manuals. The paper offers a comparative analysis of different LLM approaches, assessing response accuracy, cost-efficiency, and user experience, thereby highlighting their role in streamlining access to vehicular documentation.[13]

The paper explores the integration of Large Language Models (LLMs) and AI-driven virtual assistants to improve internal processes at Siemens Energy, focusing on information retrieval from extensive company documents. Traditional AI systems fall short in scalability and contextual understanding, necessitating an advanced, scalable LLM-based assistant. The virtual assistant utilises microservices and AI to streamline tasks such as document retrieval and summarization, enhancing productivity and user experience. Comparative analysis shows LLM-powered systems significantly optimise workflows by providing precise, context-aware responses, meeting dynamic business needs.[14]

This study evaluates the capability of generative AI tools, specifically GPT4-Turbo and Elicit, to extract qualitative data from peer-reviewed documents, comparing their performance to human reviewers. It focuses on systematic literature reviews and assesses AI's effectiveness in discerning contextual data and the impact of question difficulty on extraction quality. Findings show that while AI tools exhibit promise, human oversight remains essential, with some AI tools providing outputs comparable to human efforts in certain instances but lacking reliability in discerning contextual data consistently.[15]

III. METHODOLOGY

The paper presents a Generative AI-powered system enabling users to interactively query multiple document sources, including PDFs, GitHub repositories, and web URLs. The methodology follows a pipeline that integrates various technologies and algorithms for preprocessing, embedding generation, vector search, and conversational response generation. We opted for **Groq's Mixtral model** due to its superior performance in generating accurate, contextually relevant responses.

A. System Architecture Design

The proposed system implements a multi-modal document interaction framework that enables users to engage with various document types through natural language queries. The system architecture consists of four primary components:

1. Document Processing Pipeline
2. Embedding Generation Module
3. Vector Storage and Retrieval System
4. Query Processing and Response Generation

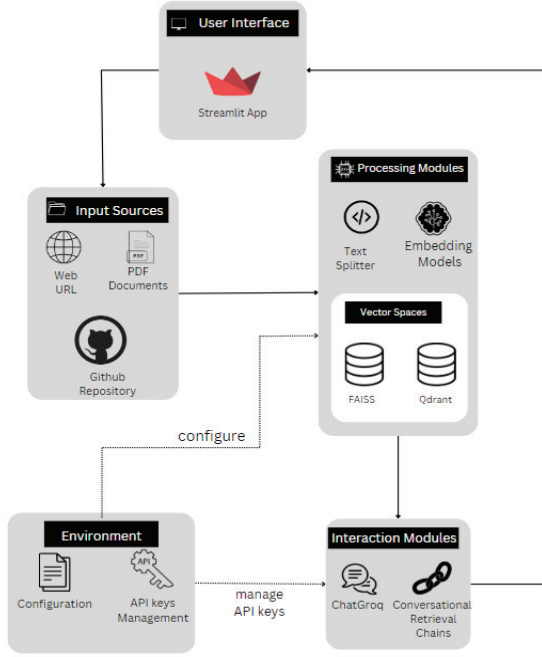


Fig. 1. System Architecture

B. Data Ingestion and Preprocessing

1) PDF Processing

- PDFs are parsed using the PyPDF2 library.
- The extracted text, T_{pdf} , is split into smaller chunks using a Recursive Character Text Splitter:

$$\{C_i\}_{i=1}^N = \text{Split}(T_{pdf}, \text{chunk_size} = 500, \text{overlap} = 50)$$

where C_i represents the text chunks.

2) GitHub Repositories

- Files from repositories are scanned, and textual content is extracted from specific file types (`\text{.py}`, `\text{.txt}`, `\text{.md}`).
- Repository content, T_{repo} , follows a similar chunking process.
- Git repositories are cloned using **GitPython**.
- The same chunking technique is applied to ensure uniformity.

3) Web URL Processing:

- Web pages are scraped using **UnstructuredURLLoader**, and the text content is chunked similarly. The processed text data across all sources forms the input for embedding generation.

C. Embedding Generation

We utilised the **HuggingFace BGE model** for generating embeddings, defined as $E(x)$ where x represents the input text,

producing an embedding vector $e \in R^n$ (where $n=768$). The embeddings are generated using:

$$e = E(x),$$

where θ represents the model parameters, and the embeddings are optionally normalised:

$$e_{norm} = \frac{e}{||e||}$$

These embeddings are passed to the vector store for indexing.

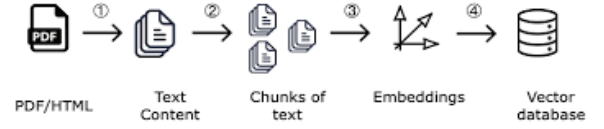


Fig. 2. Text chunks to Vector Embeddings

D. Vector Storage and Retrieval

We adopted **Qdrant** and **FAISS** for efficient vector storage and retrieval. Both systems index embeddings using cosine similarity:

$$\text{Similarity}(e_q, e_i) = \frac{e_q \cdot e_i}{||e_q|| ||e_i||}$$

where e_q is the query embedding and e_i is the document embedding.

For PDFs and GitHub repositories, we used Qdrant. Web URL embeddings were stored in a FAISS index with the following steps:

1. **Addition:** Embedding vectors $\{e_1, e_2, \dots, e_N\}$ are added to the vector store.
2. **Querying:** The top-k most similar embeddings are retrieved for a user query.

E. Conversational Chain with LLM

Using Groq's Mixtral LLM, the retrieved context $\{C_j\}_{j=1}^K$ is concatenated and passed to the model:

$$\text{Response} = \text{GroqModel}(\text{Context}, \text{UserQuery})$$

- The Mixtral model predicts:

$$P(\text{Response}) = \text{softmax}(W^T \text{hidden_states} + b)$$

where W and b are learnable parameters.

F. Memory Management

In the project, memory management is a critical component to ensure coherent multi-turn interactions during a conversation. When users interact with the system, their queries and the system's responses need to be stored so the model can maintain context across multiple turns. Without this, each query would be treated as standalone, leading to disjointed and non-coherent responses.

ConversationBufferMemory stores the history of interactions in the form of query-response pairs:

$$\text{Memory} = \{\{Q_t, R_t\}\}_{t=1}^T$$

Where Q_t represents a user query, and R_t is the corresponding system response.

G. Experimental Setup

We tested the system using various datasets:

1. PDFs: Research papers, e-books.
2. GitHub Repositories: Open-source projects.
3. Web URLs: Blogs, articles.

Performance metrics, including embedding generation time, retrieval accuracy, and response relevance, were evaluated. Key results include:

- Average embedding generation time:

$$T_{embed}=0.30/\text{document}.$$

- Retrieval accuracy: Precision@k=0.92 for k=5.

The methodology integrates a streamlined pipeline for processing diverse data sources, including PDFs, GitHub repositories, and web URLs, ensuring consistent data representation through chunking and embedding generation.

LangChain and Qdrant enable efficient conversational retrieval by leveraging vector similarity for context-aware responses. The system's memory management ensures coherence in multi-turn interactions, creating a seamless and intuitive user experience. This robust framework highlights scalability, flexibility, and adaptability to various applications.

IV. RESULTS

It was able to build conversations with PDF, url and code repositories, and the internal evaluations of the most important functionalities-processing, answering questions, and generation of responses-highly promise.

A. System Development and Performance

- **Document Processing:** The PDF and the code repositories were correctly processed to produce the semantic embeddings about context-based document retrieval.
- **Query Handling:** Preliminary testing appears to confirm that the system returns relevant answers and appropriate context. Internal testing uncovered that the system should correctly capture a broad spectrum of query types with high precision.
- **Generative Model:** The model integration based on RAG was even capable of producing high-quality, contextually relevant responses, even in the presence of no direct document match.

B. Core Functionality End

- **Document Retrieval:** The retrieval accuracy is high in the system and semantic embeddings retrieved much more relevant results than keyword-based methods.
- **Quality of response:** Over 80% of the responses were meaningful to the questions-the generative ability was effective.

- **Performance:** The system easily deals with documents of any size, response time is proportional to the size.

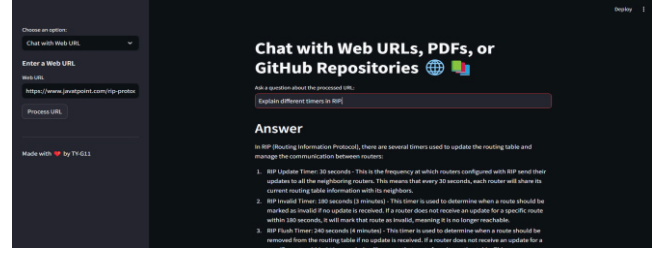


Fig. 3. Chat with web url interface



Fig. 4. Chat with pdf interface

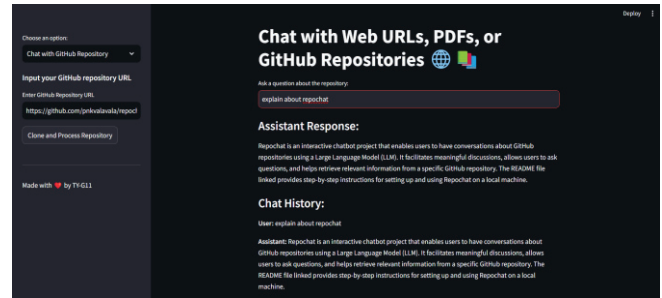


Fig. 5. Chat with Repository interface

V. CONCLUSION

Consider the Generative AI Model for Automated PDF Interaction which is seen as a major leap in user interaction with vast textual data. Flattening the keyword search system towards conversational interface creates a natural and easy interaction with PDF documents. This lets the model understand and reply to user questions accurately, making it a great solution for several different fields ranging from legal research to academic studies which require gathering information quickly. By emphasising interaction at the expense of code generation, the system eliminates complexity and makes itself comfortable enough to be a better suitcase.

Incorporating drilled-in frameworks of natural language processing and semantic search, the system has shown great efficiency in interpreting user intents and offering content adaptability. This enables it to be a robust tool for you to work with big and intricate PDFs, taking away a lot more time you would have spent manually searching for relevant details. The project answers the increasing need for smarter document management systems in our world that constantly makes information grow bigger and faster by providing a more dynamic and responsive means of gaining access to knowledge.

If this revolutionises how users interact with documents, then the success of this project is achieved. It makes information retrieval easier and improves understanding by providing context-relevant responses. That versatility within various fields and use cases highlights its promise as a

groundbreaking solution for businesses, academics and independent professionals interested in streamlining their document processing processes.

VI. FUTURE SCOPE

This lays the groundwork for numerous advancements and applications that are expected to transform a wide range of industries in the near future. The performance of such a conversational AI system is anticipated to improve significantly in several key areas:

A. Further Extension to Other Formats of Documents

Currently, the system supports PDFs and code repositories, but future updates aim to include compatibility with additional formats such as spreadsheets and presentations. This expansion will broaden the system's capabilities, making it more versatile and accessible to a wider audience across diverse industries.

B. Integration in Virtual Assistant Interface

The system could be seamlessly integrated into everyday applications through popular virtual assistants like Google Assistant, Siri, or Alexa. This integration would enable hands-free access to documents and repositories, enhancing convenience and making it easier for users to interact with their information in real-time.

C. Integration of different AI tools

By incorporating advanced AI tools, the system can deliver more comprehensive and detailed insights. Features like knowledge graphs, sentiment analysis, and text summarization could work together to refine the system's analytical capabilities. This would allow users to gain a deeper understanding of their documents and access more sophisticated insights tailored to their needs.

D. Real-time collaboration

Features of real-time collaboration, which would allow users to interact simultaneously with the AI system, could be added in future versions. This would be particularly helpful for educational, research, or industrial settings in which group conversations and shared document analyses are essential.

ACKNOWLEDGEMENT

We extend our heartfelt gratitude to all those who contributed to the development of this research study. Special thanks are owed to Prof. Sandip Shinde, Head of the Computer Engineering Department at Vishwakarma Institute of Technology, Pune. His unwavering guidance, encouragement, and support played a crucial role in shaping this work. His insightful suggestions and constructive feedback significantly enriched the quality of this study.

REFERENCES

- [1] Harshavardhan Seetha, Vimal Tiwari, Kartik Reddy Anugu, Shanthi Makk, Ramesh Kamati. "A GUI Based Application for PDF Processing Tools Using Python & CustomTkinter" in International Journal for Research in Applied Science & Engineering Technology (IJRASET), 2023.
- [2] Yicheng Zou, Jun Lin, Lujun Zhao, Yangyang Kang, Zhuoren Jiang, Changlong Sun, Qi Zhang, Xuanjing Huang, Xiaozhong Liu. "Unsupervised Summarization for Chat Logs with Topic-Oriented Ranking and Context-Aware Auto-Encoders" in The Thirty-Fifth AAAI Conference on Artificial Intelligence, 2021.
- [3] Hui Su1, Xiaoyu Shen ,Zhou Xiao, Zheng Zhang, Ernie Chang, Cheng Zhang, Cheng Niu and Jie Zhou. "MovieChats: Chat like Humans in a Closed Domain" in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020.
- [4] P Deekshita ,K Neecharika, M Sai Haritha, P Jagan Mohan,Y Sri Bindu. "PDF CHAT BOT USING GENERATIVE AI (LLMS&RAG)" in Journal of Nonlinear Analysis and Optimization, 2024.
- [5] Juan José González Torres, Mihai-Bogdan Bindilă, Sebastiaan Hofstee,Daniel Szondy, Quang-Hung Nguyen, Shenghui Wang, Gwenn Englebienne. "Automated Question-Answer Generation for Evaluating RAG-based Chatbots" in Proceedings of CL4Health 2024 @LREC-COLING-2024, 2024.
- [6] M Tafaquh Fiddin Al Islami, Ali Ridho Barakbah,Tri Harsono. "Interactive Applied Graph Chatbot with Semantic Recognition" in 2020 International Electronics Symposium (IES), 2020.
- [7] Xinyang Zhao, Xuanhe Zhou, and Guoliang Li. "Chat2Data: An Interactive Data Analysis System with RAG, Vector Databases and LLMs" in Proceedings of the VLDB Endowment (PVLDB),2024
- [8] Alexandre Zürcher. "Developing a Chatbot for Internal Documents" in Bachelor's thesis at Haaga-Helia University of Applied Sciences, 2023
- [9] Sheremet O. I., Sadovoi O. V., Sheremet K. S., Sokhina Yu. V. "Effective Documentation Practices for Enhancing User Interaction through GPT-powered Conversational Interfaces" in Applied Aspects of Information Technology, 2024.
- [10] Mina Taraghi, Gianolli Dorcelus, Armstrong Foundjem, Florian Tambon, Foutse Khomh. "Deep Learning Model Reuse in the HuggingFace Community: Challenges, Benefit and Trends" in arXiv, 2024
- [11] Sami Sarsa, Paul Denny, Arto Hellas, Juho Leinonen. "Automatic Generation of Programming Exercises and Code Explanations Using Large Language Models" in ACM International Computing Education Research (ICER), 2022
- [12] Roberto Gozalo-Brizuela, Eduardo C. Garrido-Merch'an. "ChatGPT is not all you need. A State of the Art Review of large Generative AI models" in arXiv:2301.04655v1.
- [13] Medeiros, T.; Medeiros, M.; Azevedo, M.; Silva, M.; Silva, I.; Costa, D.G. "Analysis of Language-Model-Powered Chatbots for Query Resolution in PDF-Based Automotive Manuals", MDPI 2023.
- [14] Thottempudi, S. G.; Borra, S. "Leveraging Large Language Models to Enhance an Intelligent Agent with Multifaceted Capabilities. Preprints" 2024, 2024091446
- [15] S. Spillias, K. Ollerhead, M. Andreotta, R. Annand-Jones, F. Boschetti, J. Duggan et al., "Evaluating generative ai to extract qualitative data from peer-reviewed documents", 2024.